

▣ 16 Descriptive Statistics with R

In the chapters on descriptive statistics we looked at various metrics for identifying central tendency and spread for variables of different types. We can summarise these as follows.

Measures of average			
	Mean	Median	Mode
Scale	✓	✓	✓
Ordinal		✓	✓
Nominal			✓
Binary			✓

Measures of spread				
	Standard deviation (SD)	Range	IQR or semi-IQR	No of levels
Scale	✓	✓	✓	
Ordinal			✓	✓
Nominal				✓

We also looked at a variety of charts for a single variable and when we might use them – a summary of these are given below.

Variable Type	Chart type
Scale continuous or discrete	histogram, stem and leaf plot, boxplot or dotplot
Categorical ordinal, nominal or binary	bar chart or pie chart

✎ *You would be ill-advised to blindly follow the guides in the table above. A statistic or chart might, in theory, be an option for a particular purpose but it may not prove to be a good choice. A question you should ask in relation to any such construct is – have I best conveyed the information in the data?*

Here we will apply this to an actual data set - the customer survey data.

The survey instrument used to generate the data can be viewed [here](#) and the full data set in CSV format can be obtained from [here](#).

The first three questions along with the data for the first five customers is shown below.

Q1 What is your gender?

Male ☐ Female ☐

Q2 What age were you on your last birthday?

Q3 What type of goods did you purchase (tick one only)?

electronics ☐ jewellery ☐ fragrances ☐

q1_sex	q2_age	q3_goods	
2	59	2	
2	32	2	
2	62	2	
1	41	1	
1	38	NA	
:	:	:	

Note that categorical variables are coded with integers (i.e. 1, 2, 3) and in the order in which they appear on the survey (e.g. Male $\Leftrightarrow$ 1, Female $\Leftrightarrow$ 2)

Q1 What is your gender?

Male ☐ Female ☐

Loading the data

Assume the data is stored in a csv delimited text file called “customers.csv” located in the directory “C:\stats\data”. To load the data, it is first necessary to navigate to the appropriate working directory and load the data set and store it in a suitable dataframe variable . i.e.

```
setwd("C://stats//data")
dir() # check dir contents
customers = read.table("customers.csv", sep= ",", header = TRUE)
```

Copy and paste using the clipboard

Often data will be already compiled in some windows application such as MS Excel. In such cases the data may be copied and pasted from such applications to R. First the data should be selected and copied (CTRL + C) then

```
customers = read.table("clipboard", header = T)
```

Note that TRUE may be abbreviated to T and similarly F can be used for FALSE (provided you have not used T and F as variables)

Viewing the data

There are a number of R constructs that may be used to view all, or part of the data.

```
customers          # Show all of the data. Scrolls off screen!
head(customers)    # Show the first few rows
names(customers)   # Show the variable names
customers [1,1]    # Show the age of the first customer
customers [1,]     # Show all data for the first variable (col 1)
customers [,1]     # Show all data for the first customer (row 1)
customers [1:3,]   # Show the data for the first three customer
customers [,1:3]   # Show data for the first 3 variables
customers [1:3,1:5] # Show data for the first 3 vars & 5 customers
edit(customers)    # View data in an editable GUI
```

Exploring a scale variable

Age (q2_age)

```
mean(customers$q2_age)          # mean
summary(customers$q2_age)       # mean and 5 point summary
sd(customers$q2_age)            # standard deviation
```

Exercise

Investigate how well the rule of thumb, $s = \frac{1}{4}$ range works here. Also, find the semi-IQR and assess how good an approximation it is for the standard deviation. (Why might you expect these to be similar?)

Attach

Having to proceed the variable with customers\$ can get tedious. We can access the variables directly by attaching the dataframe to the search path of R

```
search()
attach(customers)
search()
```

Be careful when using attach. A variable called q2_age can be defined at the command prompt and if it has, then it will be this variable that is invoked if we type q2_age after attaching customers

```
q2_age = 4          # defined at the R command prompt
customers$q2_age    # age variable in the customers dataframe
q2_age             # this will yield the value 4, even after attach
rm(q2_age)          # delete the variable defined at the prompt
q2_age             # this will yield the age variable data
```

From here on, we assume that the dataframe customers is attached. But you can remove a dataframe from the search path with

```
detach(customers)
```

Note that we can generate charts of the age variables as follows

```
hist(q2_age)
stem(q2_age)
boxplot(q2_age)
```

Exercise

Can you guess what the additional arguments below to boxplot and hist do?

```
hist(q2_age, xlab="Age", main = "")
boxplot(q2_age, horizontal=TRUE)
```

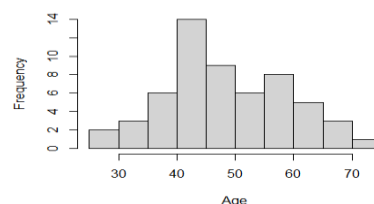
Spend (q4_spend)

We could similarly analyse spend – but we encounter a problem because there is some missing data (NA)

```
mean(q4_spend)           # returns NA
mean(q4_spend, na.rm = TRUE) # returns 55.90536
```

Summary of age

The customers profile for this site tends to be older (mean 48.8, SD = 10.6 years). Indeed, as the plot opposite shows, the youngest customer in the sample was in their late twenties.



Exercise Analyse spend and delivery time

Exploring a nominal variable

Goods (q3_goods) Remember electronics = 1, jewellery =2, fragrances =3

```
q3_goods           # view all the data - note there are 2 NA
mean(q3_goods, na.rm=TRUE) # yields 1.96 - meaningless statistic!
table(q3_goods)      # yields frequencies of each goods type
```

```
q3_goods
 1  2  3
16 25 14
sum(table(q3_goods)) # Yields 55 (remember there's 57 customers)
table(q3_goods)/55   # yields percentages of each goods type
```

```
q3_goods
      1      2      3
0.2909091 0.4545455 0.2545455
```

```
round(table(q3_goods)/55,2) # Round the proportions to 2 DP
```

```
q3_goods
 1  2  3
0.29 0.45 0.25
```

Thus, in our sample, 29% buy electronics, 45% jewellery and 25% fragrances.

```
barplot(table(q3_goods))          # bar chart of goods
```

Exercise

Convince yourself that the following lines will yield the percentage, frequency bar chart shown

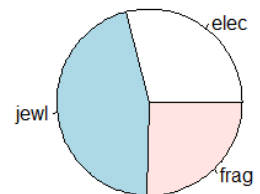
```
goods = table(q3_goods)
goodsTotal = sum(goods)
names(goods) <- c("elec", "jewl", "frag")
barplot(goods/total, main = "Goods sold (% freq)")
```



Exercise What do you suppose `pie(goods)` will generate?

Summary of goods

While jewellery is the most popular product type, accounting for 45% of goods sold, sales of electronics and fragrances are also considerable, accounting for 29% and 25% of the total respectively.



Notes

1. In effect we have reported here that the mode for goods is jewellery – but we have not used the term “mode” in our report.
2. A similar analyses could be carried out on a binary variable – for example, we might report that there were more females (31) than males (25).

Exploring an ordinal variable

Concern about online fraud (*q10_fraudRisk*)

Q10 When shopping online, what is your view of the risk of online fraud?

Very risky ☐ A little risky ☐ No opinion ☐ Fairly safe ☐ Very safe ☐

```
> summary(q10_fraudRisk)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's 
    1      2      3      3      4      5      1 
> table(q10_fraudRisk)
q10_fraudRisk
 1  2  3  4  5 
 6 12 21 10  7
```

It is doubtful if reporting any measure of central tendency or spread will be very informative here. The median and mode are both 3 (the mean is meaningless). Even though the variable is ordinal, because there are only five distinct levels, reporting quartiles / IQR or the number of distinct levels is not going to be very informative. The symmetry evident in the tabular output would suggest that respondents are not leaning towards one end of the scale or other in their attitude to fraud risk. The summary of this variable might reflect this without explicitly reporting any measure of central tendency or spread.

Summary of concern about online fraud

Respondents are equally split in their attitude to concern about online fraud with 18/56 (32%) indicating that the consider it a little, or very risky and 17/56 (30%) indicating that consider it fairly or very safe.